Jio Platforms Limited

NGINX LUA GUIDE

Consent Management System

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Version History

Date	Contributor	Contribution	Version	State
Nov 25, 2025	Racharla Vamshi	Initial Draft	0.1	Draft
Nov 26, 2025	Lokesh Sharma	Reviewed and released the document	1.0	Released

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1. Download and extract the OpenResty source:

- wget https://openresty.org/download/openresty-VERSION.tar.gz
  tar -xzf openresty-VERSION.tar.gz
  cd openresty-VERSION
- recommended version & tested : 1.25.3.2

2. Create a portable directory structure:

```
mkdir -p /app/build
mkdir -p /app/lua/luajit/lib
mkdir -p /app/lua/luajit/include
cd /app/build
```

3. Copy LuaJIT libraries and headers into the portable directory if these files are not there install them using `sudo yum install luajit luajit-devel`:

```
cp /usr/lib64/libluajit-5.1.so* /app/lua/luajit/lib/
cp -r /usr/include/luajit-2.1/* /app/lua/luajit/include/
```

4. (Optional) Download extra modules if required

```
cd /app/build
git clone https://github.com/openresty/headers-more-nginx-module
git clone https://github.com/openresty/echo-nginx-module
git clone https://github.com/openresty/redis-nginx-module
```

5. Configure OpenResty to use the portable LuaJIT paths:

```
cd /app/build/openresty-1.25.3.2

./configure \
--prefix=/app/lua/nginx \
--with-http_ssl_module \
--with-http_v2_module \
--with-http_stub_status_module \
--with-stream \
--with-stream_ssl_module \
--with-stream_ssl_preread_module \
--with-pcre-jit \
--with-luajit \
--with-cc-opt="-O2 -I/app/lua/luajit/include" \
--with-ld-opt="-L/app/lua/luajit/lib -Wl,-rpath,/app/lua/luajit/lib"
(If external modules are needed, add them like:)
--add-module=/app/build/headers-more-nginx-module
```

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6. Build and install:

```
make -j$(nproc)
make install
```

7. Verified Build Information:

The Nginx-Lua setup was tested using the following build:

Command:
/app/lua/nginx/sbin/nginx -V

Output:

```
nginx version: openresty/1.25.3.2
built by gcc 11.4.1 (Red Hat 11.4.1-4)
built with OpenSSL 3.0.7 (running with OpenSSL 3.2.2)
TLS SNI support enabled

configure arguments:
--prefix=/app/lua/nginx
--with-cc-opt='-O2 -I/app/lua/luajit/include'
--with-ld-opt='-Wl,-rpath,/app/lua/luajit/lib -L/app/lua/luajit/lib -Wl,-
rpath,/app/lua/luajit/lib'
--add-module=../ngx_devel_kit-0.3.3
--add-module=../echo-nginx-module-0.63
--add-module=../xss-nginx-module-0.06
--add-module=../ngx_coolkit-0.2
--add-module=../set-misc-nginx-module-0.33
--add-module=../form-input-nginx-module-0.12
--add-module=../encrypted-session-nginx-module-0.09
--add-module=../srcache-nginx-module-0.33
--add-module=../ngx_lua-0.10.26
--add-module=../ngx_lua_upstream-0.07
--add-module=../headers-more-nginx-module-0.37
--add-module=../array-var-nginx-module-0.06
--add-module=../memc-nginx-module-0.20
--add-module=../redis2-nginx-module-0.15
--add-module=../redis-nginx-module-0.3.9
--add-module=../rds-json-nginx-module-0.16
--add-module=../rds-csv-nginx-module-0.09
--add-module=../ngx_stream_lua-0.0.14
--with-http_ssl_module
--with-http_v2_module
--with-http_stub_status_module
--with-pcre-jit
--with-stream
```

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--without-pcre2
--with-stream_ssl_module
--with-stream_ssl_preread_module

8. Kill any existing nginx instances if needed:

ps -ef | grep nginx
kill -9 <pid>

9. Start the portable NGINX:

./nginx

10. Package the portable build:

cd /app
zip -r nginx-lua.zip lua

11. Result:

The nginx-lua build will be located in /app/lua/nginx and can be copied to another compatible Linux machine.

Configuration:

This configuration is similar to a standard Nginx configuration, but it includes some additional directives and modules specific to the Nginx-Lua setup.

Example server and location configurations are shown below. :

```
server {
    listen 8443 ssl;
    server_name api.negd.com;

    ssl_certificate     /app/middleware/nginx/cert/jcms.negd.pem;
    ssl_certificate_key
/app/middleware/nginx/cert/jcms.negd.com.key;

    server_tokens off;

    ssl_session_cache shared:SSL:1m;
    ssl_session_timeout 10m;
    ssl_protocols TLSv1.2;
    ssl_ciphers 'ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES256-GCM-
SHA384:ECDHE-ECDSA-CHACHA20-POLY1305:ECDHE-RSA-CHACHA20-
POLY1305:ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA-AES128-GCM-SHA256';
```

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```
    ssl_prefer_server_ciphers on;

    access_log /app/lua/nginx/logs/ip_debug.log ipdebug;

    location /test {
        default_type text/plain;
        content_by_lua_block {
            ngx.say("Nginx Lua works")
        }
    }

    location /oauth2/ {
        proxy_set_header X-Forwarded-Host $host;
        proxy_set_header X-Forwarded-Server $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $http_host;
        proxy_read_timeout 5m;
        proxy_send_timeout 5m;
        proxy_hide_header X-Forwarded-Host;
        proxy_hide_header X-Forwarded-Server;
        proxy_hide_header X-Forwarded-For;
        proxy_hide_header Referrer-Policy;
        proxy_hide_header X-Content-Type-Options;
        proxy_hide_header X-Frame-Options;
        proxy_hide_header X-XSS-Protection;
        proxy_hide_header Content-Security-Policy;
        proxy_hide_header Strict-Transport-Security;

        proxy_pass http://jcms.auth.com/oauth2/;
        allow 127.0.0.1;
        deny all;
    }

    location /secureCode/ {
        proxy_set_header X-Forwarded-Host $host;
        proxy_set_header X-Forwarded-Server $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $http_host;
        proxy_read_timeout 5m;
        proxy_send_timeout 5m;
        proxy_hide_header X-Forwarded-Host;
        proxy_hide_header X-Forwarded-Server;
        proxy_hide_header X-Forwarded-For;
        proxy_hide_header Referrer-Policy;
        proxy_hide_header X-Content-Type-Options;
        proxy_hide_header X-Frame-Options;
        proxy_hide_header X-XSS-Protection;
```

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```
        proxy_hide_header Content-Security-Policy;
        proxy_hide_header Strict-Transport-Security;

        proxy_pass http://jcms.auth.com/secureCode/;
        allow 127.0.0.1;
        deny all;
    }

    location /api/securecode/ {

        if ($request_method = OPTIONS) {
            add_header 'Access-Control-Allow-Origin' '*';
            add_header 'Access-Control-Allow-Methods' 'GET, POST,
PUT, PATCH, DELETE, OPTIONS';
            add_header 'Access-Control-Allow-Headers' '*';
            add_header 'Access-Control-Max-Age' 86400;
            add_header 'Content-Length' 0;
            add_header 'Content-Type' 'text/plain';
            return 204;
        }

        proxy_set_header X-Forwarded-Host $host;
        proxy_set_header X-Forwarded-Server $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $http_host;
        proxy_read_timeout 5m;
        proxy_send_timeout 5m;
        proxy_hide_header X-Forwarded-Host;
        proxy_hide_header X-Forwarded-Server;
        proxy_hide_header X-Forwarded-For;
        proxy_hide_header Referrer-Policy;
        proxy_hide_header X-Content-Type-Options;
        proxy_hide_header X-Frame-Options;
        proxy_hide_header X-XSS-Protection;
        proxy_hide_header Content-Security-Policy;
        proxy_hide_header Strict-Transport-Security;
        proxy_pass https://jcms.api.gateway/secureCode/v1.0/;
        allow all;
    }

    location /token {
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;
        proxy_set_header X-Forwarded-Host $host;
        proxy_set_header X-Forwarded-Server $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $http_host;
```

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```
        proxy_read_timeout 5m;
        proxy_send_timeout 5m;

        proxy_hide_header X-Forwarded-Host;
        proxy_hide_header X-Forwarded-Server;
        proxy_hide_header X-Forwarded-For;
        proxy_hide_header Referrer-Policy;
        proxy_hide_header X-Content-Type-Options;
        proxy_hide_header X-Frame-Options;
        proxy_hide_header X-XSS-Protection;
        proxy_hide_header Content-Security-Policy;
        proxy_hide_header Strict-Transport-Security;

        add_header 'Access-Control-Allow-Origin' '*';
        add_header 'Access-Control-Allow-Methods' 'GET, POST, PUT,
PATCH, DELETE, OPTIONS';
        add_header 'Access-Control-Allow-Headers' '*';
        add_header 'Access-Control-Max-Age' 86400;
        proxy_pass https://jcms.token.gateway/oauth2/token;
        allow all;
    }

    location /schedular/ {
        access_by_lua_file
/app/lua/nginx/conf.d/lua/partnerportal.lua;
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;
        proxy_set_header X-User $http_x_user;
        proxy_set_header Authorization $http_authorization;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;
        proxy_pass https://jcms.api.gateway/scheduler/v1.0/;
    }

    location /cookie/ {
        access_by_lua_file
/app/lua/nginx/conf.d/lua/partnerportal.lua;
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;
        proxy_set_header X-User $http_x_user;
        proxy_set_header Authorization $http_authorization;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;
        proxy_pass https://jcms.api.gateway/cookie/v1.0/;
    }

    location /notification/ {
        access_by_lua_file
/app/lua/nginx/conf.d/lua/partnerportal.lua;
```

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```
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;
        proxy_set_header X-User $http_x_user;
        proxy_set_header Authorization $http_authorization;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;
        proxy_pass https://jcms.api.gateway/notification/v1.0/;
    }

    location /grievance/ {
        access_by_lua_file
/app/lua/nginx/conf.d/lua/partnerportal.lua;
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;
        proxy_set_header X-User $http_x_user;
        proxy_set_header Authorization $http_authorization;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;
        proxy_pass https://jcms.api.gateway/grievance/v1.0/;
    }

    location /translator/ {
        access_by_lua_file
/app/lua/nginx/conf.d/lua/partnerportal.lua;
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;
        proxy_set_header X-User $http_x_user;
        proxy_set_header Authorization $http_authorization;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;
        proxy_pass https://jcms.api.gateway/translator/v1.0/;
    }

    location /consent/ {
        access_by_lua_file
/app/lua/nginx/conf.d/lua/partnerportal.lua;
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;
        proxy_set_header X-User $http_x_user;
        proxy_set_header Authorization $http_authorization;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;
        proxy_pass https://jcms.api.gateway/consent/v1.0/;
    }

    location /audit/ {
        access_by_lua_file
/app/lua/nginx/conf.d/lua/partnerportal.lua;
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;
        proxy_set_header X-User $http_x_user;
        proxy_set_header Authorization $http_authorization;
```

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```
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;
        proxy_pass https://jcms.api.gateway/audit/v1.0/;
    }
    location /partnerportal/ {
        access_by_lua_file
/app/lua/nginx/conf.d/lua/partnerportal.lua;
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;
        proxy_set_header X-User $http_x_user;
        proxy_set_header Authorization $http_authorization;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;
        proxy_pass https://jcms.api.gateway/partnerportal/v1.0/;
    }

    location /scheduler/ {
        access_by_lua_file
/app/lua/nginx/conf.d/lua/partnerportal.lua;
        include /app/lua/nginx/conf.d/cors-config/preflight.conf;

        proxy_set_header X-User $http_x_user;
        proxy_set_header Authorization $http_authorauthorization;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;

        proxy_pass https://jcms.api.gateway/scheduler/v1.0/;
    }

    include /app/lua/nginx/conf.d/locations/user.portal.conf;
    include /app/lua/nginx/conf.d/locations/unsecure.partner.conf;
}
```

Example-Lua use for Partner Portal: partner-portal.lua

```
local http = require "resty.http"
local cjson = require "cjson.safe"
-- get token from incoming headers
local headers = ngx.req.get_headers()
local auth_header = headers["Authorization"] or headers["x-session-
token"] or headers["X-Session-Token"]
local businessId = headers["businessId"] or headers["businessid"] or
headers["business-id"]
if not auth_header then
    return ngx.exit(401)
end
-- extract token
local _, _, token = string.find(auth_header, "Bearer%s+(.+)")
```

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```
if not token then
    token = auth_header
end
if not token or token == "" then
    return ngx.exit(401)
end
-- make HTTP request to introspection endpoint
local httpc = http.new()
local body = ngx.encode_args({ token = token })
local res, err =
httpc:request_uri("https://127.0.0.1:8443/oauth2/validate", {
    method = "POST",
    body = body,
    headers = {
        ["Content-Type"] = "application/x-www-form-urlencoded",
    },
    ssl_verify = false
})
if not res then
    ngx.log(ngx.ERR, "introspection failed: ", err)
    return ngx.exit(500)
end
if res.status ~= 200 then
    ngx.header["Access-Control-Allow-Origin"]  = "*"
    ngx.header["Access-Control-Allow-Methods"] = "GET, POST,
OPTIONS"
    ngx.header["Access-Control-Allow-Headers"] = "Authorization,
Content-Type"
    ngx.header["Access-Control-Max-Age"]       = "3600"
    ngx.log(ngx.ERR, "introspection http status: ", res.status)
    return ngx.exit(401)
end
-- parse JSON response
local data = cjson.decode(res.body)
if not data or not data.active then
    return ngx.exit(401)
end
-- optionally pass info upstream
if data.sub then
    ngx.req.set_header("X-User", data.sub)
end
-- ngx.req.set_header("X-Introspection", res.body)
-- set Authorization header for upstream
if data.accessToken and data.accessToken ~= "" then
    ngx.req.set_header("Authorization", "Bearer " ..
data.accessToken)
```

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```
end
```

Example userportal.lua:

```
local http = require "resty.http"
local cjson = require "cjson.safe"
local headers = ngx.req.get_headers()
local access_token = headers["x-secure-code"] or
headers["accessToken"]
if not access_token then
    ngx.log(ngx.ERR, "Missing access token in headers")
    return ngx.exit(401)
end

local httpc = http.new()
local res, err =
httpc:request_uri("https://127.0.0.1:8443/secureCode/validate", {
    method = "GET",
    headers = {
        ["accessToken"] = access_token,
    },
    keepalive = false,
    ssl_verify = false
})
if not res then
    ngx.log(ngx.ERR, "Validation request failed: ", err)
    return ngx.exit(500)
end
if res.status ~= 200 then
    ngx.log(ngx.ERR, "Validation API returned status: ", res.status)
    return ngx.exit(401)
end
local data = cjson.decode(res.body)
if not data or not data.active then
    ngx.log(ngx.ERR, "Session inactive or invalid")
    return ngx.exit(401)
end
-- Set headers for backend
ngx.req.set_header("X-Session-Active", "true")
if data.accessToken then
    ngx.req.set_header("Authorization", "Bearer " ..
data.accessToken)
else
    ngx.log(ngx.ERR, "Missing accessToken in validation response")
    return ngx.exit(500)
end
```

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a) Example tenant-auth.lua:

```
local http = require "resty.http"
local cjson = require "cjson.safe"

-- read body (required for POST/PUT)
ngx.req.read_body()
local body = ngx.req.get_body_data()

local identity = nil

if body then
    local json = cjson.decode(body)
    if json and json.spoc then
        -- pick mobile if not empty
        if json.spoc.mobile and json.spoc.mobile ~= "" then
            identity = json.spoc.mobile
        -- else pick email
        elseif json.spoc.email and json.spoc.email ~= "" then
            identity = json.spoc.email
        end
    end
end

-- read secure-code header
local headers = ngx.req.get_headers()

local secret_code = headers["Secret-Code"]
    or headers["secret-code"]
    or headers["SECURE-CODE"]

-- if identity header exists, override body logic
if not identity then
    identity = headers["identity-value"]
        or headers["Identity-Value"]
        or headers["identity-Value"]
end

-- if still not found → unauthorized
if not secret_code or not identity then
    return ngx.exit(401)
end

-- make HTTP request to introspection endpoint
local httpc = http.new()
```

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```
local res, err =
httpc:request_uri("https://127.0.0.1:8443/secureCode/tenant/validate
", {
    method = "GET",
    headers = {
        ["secret-code"] = secret_code,
        ["identityvalue"] = identity,
    },
    ssl_verify = false
})

if not res then
    ngx.log(ngx.ERR, "introspection failed: ", err)
    return ngx.exit(500)
end

if res.status ~= 200 then
    ngx.log(ngx.ERR, "introspection status: ", res.status)
    return ngx.exit(401)
end

local data = cjson.decode(res.body)
if not data or not data.active then
    return ngx.exit(401)
end
```

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